T10 Aprendizaje profundo

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1 Introducción

Aprendizaje profundo: área del ML especializada en el aprendizaje de redes con (muchas) capas ocultas

Revolución del aprendizaje profundo: el aprendizaje profundo alcanza gran popularidad a partir de los 2010, principalmente por la mejora de resultados en reconocimiento automático del habla (2011) y clasificación de imágenes (2012)

Objetivo: introducir arquitecturas destacadas del aprendizaje profundo para clasificación de imágenes principalmente

LeNet (1998): red **convolucional (CNN)** pionera, introducida en 1998 por Yann LeCun para MNIST

- Propone el uso de capas convolucionales y pooling (agrupación)

AlexNet (2012): CNN de Alex Krizhevsky para ImageNet que atrajo mucho interés

- Resultados espectaculares gracias a la profundidad, uso de GPUs, ReLUs y dropout

Inception (2014): CNN de Google que introduce la meta-arquitectura típica de las CNNs modernas

ResNet (2015): CNN de Microsoft que introduce **conexiones residuales** para entrenar modelos muy profundos

- Propone el uso de conexiones residuales y añade capas de normalización

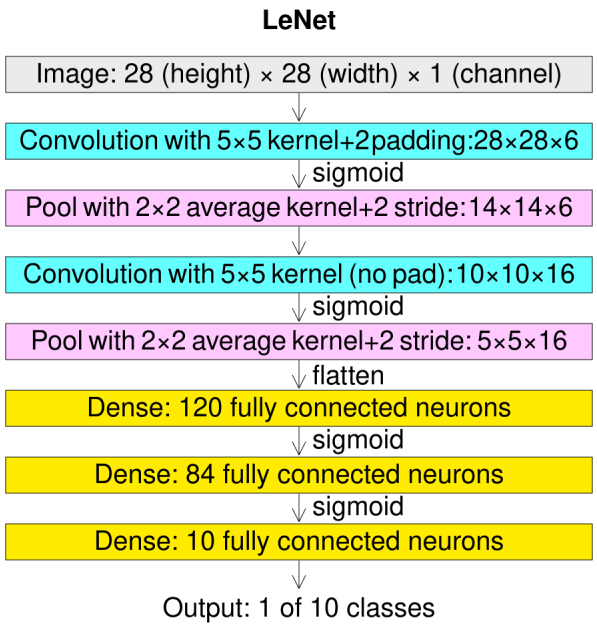
Transformer (2017): arquitectura de Google para traducción automática basada en el **mecanismo de atención**

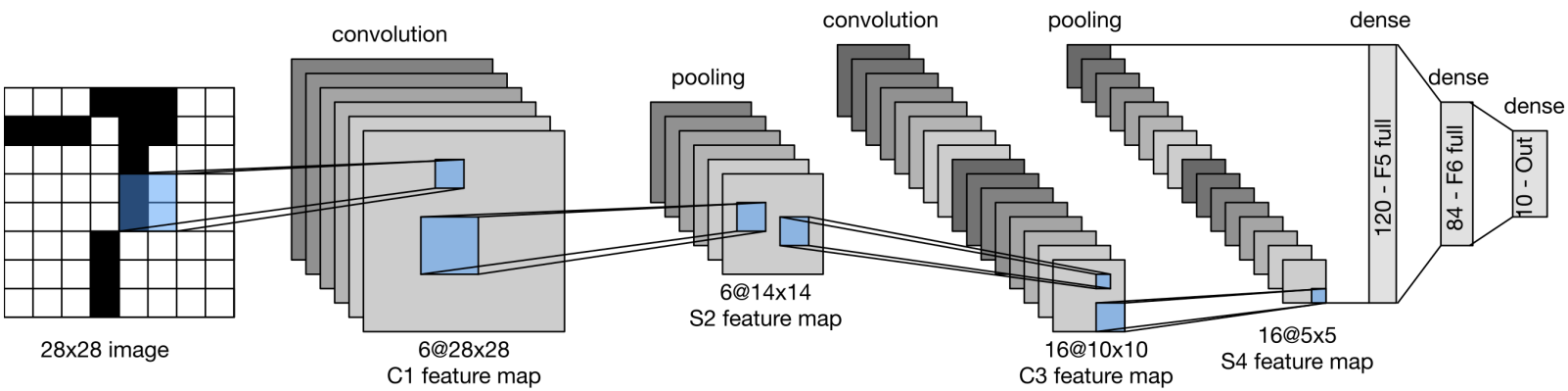
- Se considera la base de los modelos IA actuales más competitivos

Vision Transformer (2020): adaptación de Transformer para visión

2 LeNet

LeNet (1998): red **convolucional (CNN)** pionera, introducida en 1998 por Yann LeCun para MNIST





Experimento con MNIST: con relus en lugar de sigmoides

In [ ]:
```
import torch; import torch.nn as nn; import torch.nn.functional as F
class LeNet(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(in_channels=1, out_channels=6, kernel_size=5)
        self.conv2 = nn.Conv2d(in_channels=6, out_channels=16, kernel_size=5)
        self.fc1 = nn.Linear(16 * 4 * 4, 120); self.fc2 = nn.Linear(120, 84); self.fc3 = nn.Linear(84, 10)
    def forward(self, x):
        x = F.max_pool2d(F.relu(self.conv1(x)), kernel_size=2)
        x = F.max_pool2d(F.relu(self.conv2(x)), kernel_size=2)
        x = torch.flatten(x, 1); x = F.relu(self.fc1(x)); x = F.relu(self.fc2(x)); x = self.fc3(x)
        return x
device = torch.accelerator.current_accelerator().type if torch.accelerator.is_available() else "cpu"
model = LeNet().to(device); print(device, model)
```

```
cuda LeNet(
  (conv1): Conv2d(1, 6, kernel_size=(5, 5), stride=(1, 1))
  (conv2): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
  (fc1): Linear(in_features=256, out_features=120, bias=True)
  (fc2): Linear(in_features=120, out_features=84, bias=True)
  (fc3): Linear(in_features=84, out_features=10, bias=True)
)
```

In [ ]:
```
import numpy as np; import datasets
ds = datasets.load_dataset("ylecun/mnist").with_format("numpy"); ds
X_train = ds['train'][:]['image'].astype(np.float32).reshape(-1, 1, 28, 28) / 255.
y_train = ds['train'][:]['label'].astype(np.uint8)
X_test = ds['test'][:]['image'].astype(np.float32).reshape(-1, 1, 28, 28) / 255.
y_test = ds['test'][:]['label'].astype(np.uint8)
X = torch.Tensor(X_train).to(device); y = torch.Tensor(y_train).to(device, dtype=torch.long)
loss_fn = nn.CrossEntropyLoss(); optimizer = torch.optim.AdamW(model.parameters(), lr=1e-2)
model.train(); epochs = 200
for t in range(epochs):
    pred = model(X); loss = loss_fn(pred, y); loss.backward(); optimizer.step(); optimizer.zero_grad()
X = torch.Tensor(X_test).to(device); pred = model(X).detach().cpu().numpy()
acc = (pred.argmax(1) == y_test).sum().item() / y_test.size; print(f'Precisión {acc:.2%}')
```

```
Precisión 98.72%
```

3 Capas convolucionales

Valid convolution: $\mathbf{Y} = \mathbf{W} \circledast \mathbf{X}$, con un filtro $\mathbf{W}$ $f_h \times f_w$ de una imagen $\mathbf{X}$ $x_h \times x_w$

$$\mathbf{Y} = \begin{bmatrix} w_{0,0} & \cdots & w_{0,f_w-1} \\ \vdots & \ddots & \vdots \\ w_{f_h-1,0} & \cdots & w_{f_h-1,f_w-1} \end{bmatrix} \circledast \begin{bmatrix} x_{0,0} & \cdots & x_{0,x_w-1} \\ \vdots & \ddots & \vdots \\ x_{x_h-1,0} & \cdots & x_{x_h-1,x_w-1} \end{bmatrix} = \begin{bmatrix} y_{0,0} & \cdots & y_{0,y_w-1} \\ \vdots & \ddots & \vdots \\ y_{y_h-1,0} & \cdots & y_{y_h-1,y_w-1} \end{bmatrix}$$

$$y_{ij} = \sum_{u=0}^{f_h-1} \sum_{v=0}^{f_w-1} w_{u,v}\, x_{i+u,j+v} \qquad \begin{array}{ll} i = 0, \ldots, y_h - 1 & y_h = x_h - f_h + 1 \\ j = 0, \ldots, y_w - 1 & y_w = x_w - f_w + 1 \end{array}$$

Ejemplo: $f_h = f_w = 2$, $x_h = x_w = 3$, por lo que $y_h = y_w = 2$

$$\begin{bmatrix} w_{00} & w_{01} \\ w_{10} & w_{11} \end{bmatrix} \circledast \begin{bmatrix} x_{00} & x_{01} & x_{02} \\ x_{10} & x_{11} & x_{12} \\ x_{20} & x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} y_{00} & y_{01} \\ y_{10} & y_{11} \end{bmatrix}$$

$$\begin{aligned} y_{00} &= w_{00}x_{00} + w_{01}x_{01} + w_{10}x_{10} + w_{11}x_{11} \\ y_{01} &= w_{00}x_{01} + w_{01}x_{02} + w_{10}x_{11} + w_{11}x_{12} \\ y_{10} &= w_{00}x_{10} + w_{01}x_{11} + w_{10}x_{20} + w_{11}x_{21} \\ y_{11} &= w_{00}x_{11} + w_{01}x_{12} + w_{10}x_{21} + w_{11}x_{22} \end{aligned}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \circledast \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

In [ ]:
```
import numpy as np; import torch
W = torch.nn.Conv2d(1, 1, 2, bias=False)
W.weight.data = torch.tensor([[ [[1.,1.], [1.,1.]] ]], requires_grad=True)
X = torch.tensor([[ [[1,1,0], [0,1,0], [0,1,1]] ]], dtype=torch.float)
print(str(W(X).data).replace('\n',''))
```

```
tensor([[[[3., 2.],          [2., 3.]]]])
```

Zero-padding: relleno cero $2\,p_h \times 2\,p_w$, por lo que $y_h = (x_h + 2p_h - f_h + 1)$ e $y_w = (x_w + 2p_w - f_w + 1)$

Same convolution: zero-padding con $2p_h = f_h - 1$ y $2p_w = f_w - 1$ para conservar el tamaño de la entrada

Ejemplo: $f_h = f_w = 3,\ x_h = x_w = 3,\ p_h = p_w = 1$ por lo que $y_h = y_w = 3$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \circledast \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 2 \\ 4 & 5 & 4 \\ 2 & 3 & 3 \end{bmatrix}$$

In [ ]:
```
W = torch.nn.Conv2d(1, 1, 2, padding='same', bias=False)
W.weight.data = torch.tensor([[ [[1.,1.,1.], [1.,1.,1.], [1.,1.,1.]] ]], requires_grad=True)
X = torch.tensor([ [[[1,1,0], [0,1,0], [0,1,1]] ]], dtype=torch.float)
print(str(W(X).data).replace('\n',''))
```

```
tensor([[[[3., 3., 2.],          [4., 5., 4.],          [2., 3., 3.]]]])
```

Strided convolution: a saltos s_h y s_w, luego $y_h = \left\lfloor \dfrac{x_h + 2p_h - f_h + s_h}{s_h} \right\rfloor$ e $y_w = \left\lfloor \dfrac{x_w + 2p_w - f_w + s_w}{s_w} \right\rfloor$

Ejemplo: $f_h = f_w = 2,\ x_h = 2,\ x_w = 6,\ p_h = p_w = 0,\ s_h = s_w = 2,$ por lo que $y_h = 1$ e $y_w = 3$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \circledast \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 2 \end{bmatrix}$$

In [ ]:
```
W = torch.nn.Conv2d(1, 1, 2, stride=2, padding=0, bias=False)
W.weight.data = torch.tensor([[ [[1.,1.], [1.,1.]] ]], requires_grad=True)
X = torch.tensor([[ [[1,1,0,1,0,0], [0,1,0,0,1,1]] ]], dtype=torch.float)
print(str(W(X).data).replace('\n',''))
```

```
tensor([[[[3., 1., 2.]]]])
```

Múltiples canales de entrada: C canales, filtro $\mathbf{W}\ C \times f_h \times f_w$, imagen $\mathbf{X}\ C \times x_h \times x_w$

$$\mathbf{Y} = \mathbf{W} \circledast \mathbf{X} = \sum\nolimits_c \mathbf{W}_{c,:,:} \circledast \mathbf{X}_{c,:,:}$$

$$\left[\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}, \begin{bmatrix}2 & 2\\2 & 2\end{bmatrix}\right] \circledast \left[\begin{bmatrix}1 & 1 & 0\\0 & 1 & 0\\0 & 1 & 1\end{bmatrix}, \begin{bmatrix}1 & 0 & 1\\1 & 1 & 1\\0 & 0 & 1\end{bmatrix}\right] = \begin{bmatrix}9 & 8\\6 & 9\end{bmatrix}$$

In [ ]:
```
W = torch.nn.Conv2d(2, 1, 2, bias=False)
W.weight.data = torch.tensor([ [ [[1.,1.],[1.,1.]], [[2.,2.],[2.,2.]] ] ], requires_grad=True)
X = torch.tensor([ [ [[1,1,0],[0,1,0],[0,1,1]], [[1,0,1],[1,1,1],[0,0,1]] ] ], dtype=torch.float)
print(str(W(X).data).replace('\n',''))
```

```
tensor([[[[9., 8.],          [6., 9.]]]])
```

Múltiples canales de entrada y salida: C de entrada y D de salida, $\mathbf{W}\ D \times C \times f_h \times f_w$, $\mathbf{X}\ C \times x_h \times x_w$

$$\mathbf{Y}_{d,:,:} = \mathbf{W}_{d,:,:,:} \circledast \mathbf{X}$$

$$\left[\left[\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}, \begin{bmatrix}2 & 2\\2 & 2\end{bmatrix}\right], \left[\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}, \begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}\right]\right] \circledast \left[\begin{bmatrix}1 & 1 & 0\\0 & 1 & 0\\0 & 1 & 1\end{bmatrix}, \begin{bmatrix}1 & 0 & 1\\1 & 1 & 1\\0 & 0 & 1\end{bmatrix}\right] = \left[\begin{bmatrix}9 & 8\\6 & 9\end{bmatrix}, \begin{bmatrix}6 & 5\\4 & 6\end{bmatrix}\right]$$

In [ ]:
```
W = torch.nn.Conv2d(2, 2, 2, bias=False)
W.weight.data = torch.tensor([ [ [[1.,1.],[1.,1.]], [[2.,2.],[2.,2.]] ],
                               [ [[1.,1.],[1.,1.]], [[1.,1.],[1.,1.]] ] ], requires_grad=True)
X = torch.tensor([ [ [[1,1,0],[0,1,0],[0,1,1]], [[1,0,1],[1,1,1],[0,0,1]] ] ], dtype=torch.float)
print(str(W(X).data).replace('\n',''))
```

```
tensor([[[[9., 8.],          [6., 9.]],         [[6., 5.],          [4., 6.]]]])
```

Sesgos: por cada salida d puede añadirse un parámetro de sesgo b_d que se suma a cada elemento de la salida

Convolución 1x1: $f_h = f_w = 1$ para pasar de C a D canales **punto a punto**

Reducción a productos matriciales: unfolding, matrix multiplication and folding

$$\mathbf{Y} = \mathbf{W} \circledast \mathbf{X} = \mathrm{Fold}(\mathrm{Unfold}(\mathbf{X})\,\mathrm{vec}(\mathbf{W}))$$

Ejemplo: $\mathbf{X} = \left[\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \right]$ $\quad \mathbf{W} = \left[\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right]$ $\quad \mathbf{Y} = \mathbf{W} \circledast \mathbf{X} = \begin{bmatrix} 19 \\ 25 \end{bmatrix}$

$$\mathrm{Unfold}(\mathbf{X}) = \begin{bmatrix} 0 & 1 & 3 & 4 \\ 1 & 2 & 4 & 5 \end{bmatrix} \qquad \mathrm{vec}(\mathbf{W}) = \begin{bmatrix} 0 & 1 & 2 & 3 \end{bmatrix}^t \qquad \mathrm{Fold}\left(\begin{bmatrix} 19 \\ 25 \end{bmatrix} \right) = \begin{bmatrix} 19 \\ 25 \end{bmatrix}$$

In [ ]:
```python
X = torch.tensor([[ [[0], [1], [2]], [[3], [4], [5]] ]], dtype=torch.float)
print('X =', X.shape, str(X).replace('\n','').replace(' ',''))
W = torch.nn.Conv2d(2, 1, (1, 2), bias=False)
W.weight.data = torch.tensor([ [ [[0],[1]], [[2],[3]] ] ], requires_grad=True, dtype=torch.float)
print('W =', W.weight.data.shape, str(W.weight.data).replace('\n','').replace(' ',''))
YConv2d = W(X)
print('YConv2d =', YConv2d.shape, str(YConv2d).replace('\n','').replace(' ',''))
Xu = torch.nn.functional.unfold(X, kernel_size=(2, 1)).transpose(1,2) # unfold devuelve transpuesta
print('Xu =', Xu.shape, str(Xu).replace('\n','').replace(' ',''))
vecW = W.weight.data.reshape(-1, 1)
print('vec(W) =', vecW.shape, str(vecW).replace('\n','').replace(' ',''))
Yu = Xu @ vecW
print('Yu =', Yu.shape, str(Yu).replace('\n','').replace(' ',''))
Y = torch.nn.functional.fold(Yu , (2, 1), (2, 1))
print('Y =', Y.shape, str(Y).replace('\n','').replace(' ',''))
(YConv2d - Y).abs().max().item()
```

```
X = torch.Size([1, 2, 3, 1]) tensor([[[[0.],[1.],[2.]],[[3.],[4.],[5.]]]])
W = torch.Size([1, 2, 2, 1]) tensor([[[[0.],[1.]],[[2.],[3.]]]])
YConv2d = torch.Size([1, 1, 2, 1]) tensor([[[[19.],[25.]]]],grad_fn=<ConvolutionBackward0>)
Xu = torch.Size([1, 2, 4]) tensor([[[0.,1.,3.,4.],[1.,2.,4.,5.]]])
vec(W) = torch.Size([4, 1]) tensor([[0.],[1.],[2.],[3.]])
Yu = torch.Size([1, 2, 1]) tensor([[[19.],[25.]]])
Y = torch.Size([1, 1, 2, 1]) tensor([[[[19.],[25.]]]])
```

Out[ ]:
```
0.0
```

4 Pooling

Max pooling: $\mathbf{Y} = \mathrm{MaxPool2d}(\mathbf{X}),$ con un tamaño de filtro $f_h \times f_w$ de una imagen $\mathbf{X}$ $C \times x_h \times x_w$

$$\mathbf{Y}_{c,:,:} = \begin{bmatrix} y_{c,0,0} & \cdots & y_{c,0,f_w-1} \\ \vdots & \ddots & \vdots \\ y_{c,f_h-1,0} & \cdots & y_{c,f_h-1,f_w-1} \end{bmatrix} \qquad y_{cij} = \max_{u=0}^{f_h-1} \max_{v=0}^{f_w-1} x_{c,i+u,j+v} \qquad \begin{matrix} i = 0, \ldots, y_h - 1 \\ j = 0, \ldots, y_w - 1 \end{matrix} \qquad \begin{matrix} y_h = x_h - f_h \\ y_w = x_w - f_w \end{matrix}$$

Ejemplo: $\mathbf{X} = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \qquad f_h \times f_w = (2,2) \qquad \mathbf{Y} = \begin{bmatrix} 4 & 5 \\ 7 & 8 \end{bmatrix}$

In [ ]:
```
import numpy as np; import torch
P = torch.nn.MaxPool2d(2, stride=1)
X = torch.tensor([[ [[0,1,2], [3,4,5], [6,7,8]] ]], dtype=torch.float)
print(str(P(X).data).replace('\n','').replace(' ',''))
```

```
tensor([[[[4.,5.],[7.,8.]]]])
```

Average pooling: $\mathbf{Y} = \mathrm{AvgPool2d}(\mathbf{X}),$ sustituye máximo por media

Ejemplo: $\mathbf{X} = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} \qquad f_h \times f_w = (2,2) \qquad \mathbf{Y} = \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix}$

In [ ]:
```
P = torch.nn.AvgPool2d(2, stride=1)
print(str(P(X).data).replace('\n','').replace(' ',''))
```

```
tensor([[[[2.,3.],[5.,6.]]]])
```

Global max pooling:

$$\mathbf{Y}_c = \text{GlobMaxPool2d}(\mathbf{X})_c = \max_{u=0,\ldots,f_h-1} \max_{v=0,\ldots,f_w-1} \text{MaxPool2d}(\mathbf{X})_{c,u,v}$$

Ej.: $\mathbf{X} = \begin{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \end{bmatrix} \quad f_h \times f_w = (2,1) \quad \text{MaxPool2d}(\mathbf{X}) = \begin{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \end{bmatrix} \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

In [ ]:
```
P1 = torch.nn.MaxPool2d((2, 1), stride=1)
X = torch.tensor([[ [[0], [1], [2]], [[3], [4], [5]] ]], dtype=torch.float)
P1X = P1(X); print('MaxPool2d =', P1X.shape, str(P1X.data).replace('\n','').replace(' ',''))
Y, _ = P1X.squeeze(3).max(2); print('GlobMaxPool2d =', Y)
```

```
MaxPool2d = torch.Size([1, 2, 2, 1]) tensor([[[[1.],[2.]],[[4.],[5.]]]])
GlobMaxPool2d = tensor([[2., 5.]])
```

Global average pooling:

$$\mathbf{Y}_c = \text{GlobAvgPool2d}(\mathbf{X})_c = \frac{1}{f_h f_w} \sum_{u=0,\ldots,f_h-1} \sum_{v=0,\ldots,f_w-1} \text{AvgPool2d}(\mathbf{X})_{c,u,v}$$

Ej.: $\mathbf{X} = \begin{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \end{bmatrix} \quad f_h \times f_w = (2,1) \quad \text{AvgPool2d}(\mathbf{X}) = \begin{bmatrix} \begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix}, \begin{bmatrix} 3.5 \\ 4.5 \end{bmatrix} \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

In [ ]:
```
P1 = torch.nn.AvgPool2d((2, 1), stride=1); P2 = torch.nn.AdaptiveAvgPool2d(1)
X = torch.tensor([[ [[0], [1], [2]], [[3], [4], [5]] ]], dtype=torch.float)
P1X = P1(X); print('AvgPool2d =', P1X.shape, str(P1X.data).replace('\n','').replace(' ',''))
Y = P2(P1X); print('GlobAvgPool2d =', Y.shape, str(Y.data).replace('\n','').replace(' ',''))
```

```
AvgPool2d = torch.Size([1, 2, 2, 1]) tensor([[[[0.5000],[1.5000]],[[3.5000],[4.5000]]]])
GlobAvgPool2d = torch.Size([1, 2, 1, 1]) tensor([[[[1.]],[[4.]]]])
```

5 ImageNet

ImageNet: dataset de $14\,197\,122$ imágenes y $21\,841$ synsets; muy popular desde los 2010

ImageNet en Wikipedia: historia, dataset, challenge y sesgos

ImageNet en HF: ILSVRC/imagenet-1k, gmongaras/Imagenet21K, benjamin-paine/imagenet-1k-256x256, etc.

ImageNet-1K: subconjunto de 1.4M imágenes y 1000 clases

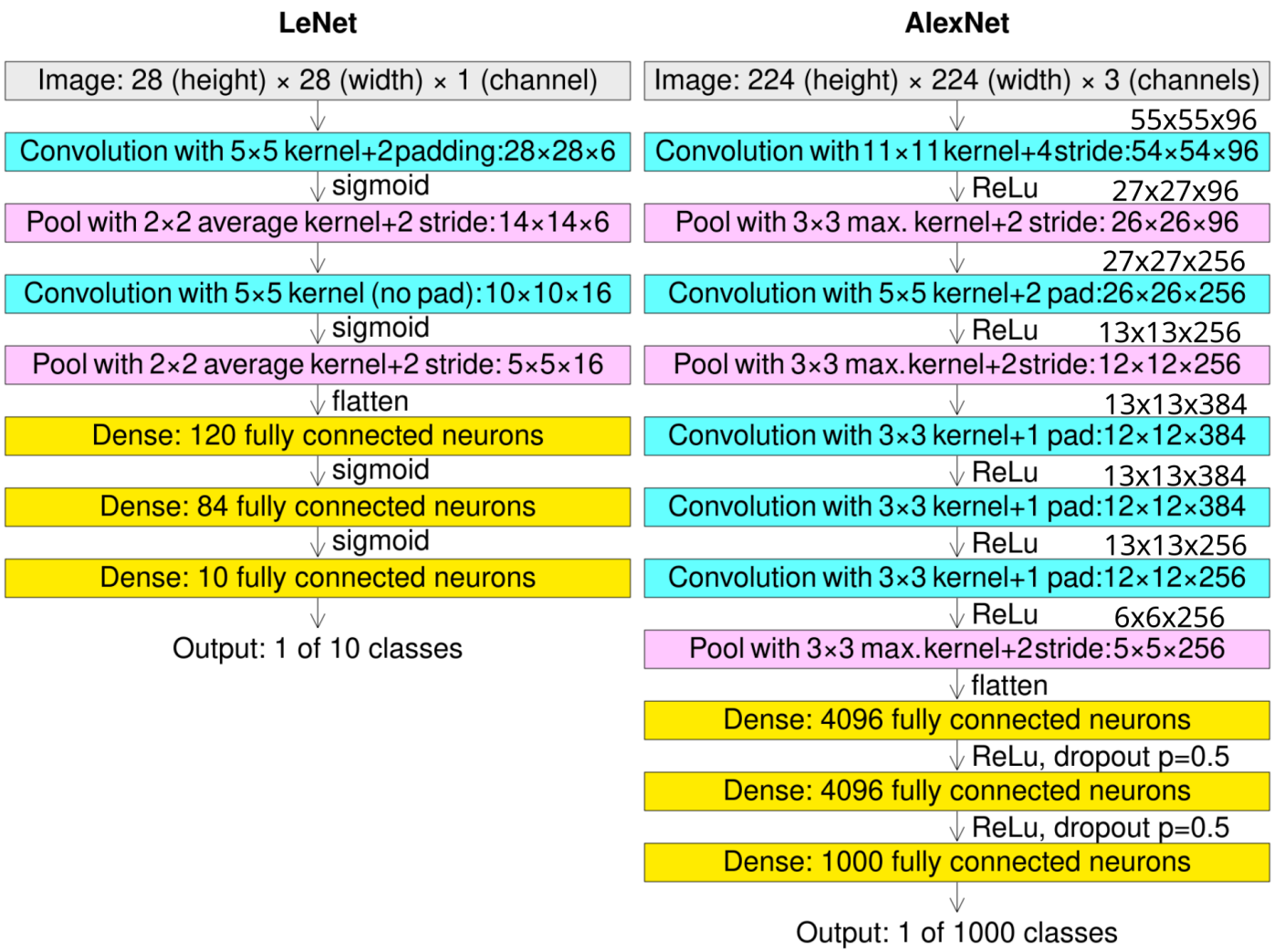
Partición estándar: $1\,281\,167$ de training, $50\,000$ de validación y $100\,000$ de test (con etiqueta de clase -1)

In [ ]:
```python
import numpy as np; import datasets
ds = datasets.load_dataset("ILSVRC/imagenet-1k"); ds
```

Out[ ]:
```
DatasetDict({
    train: Dataset({
        features: ['image', 'label'],
        num_rows: 1281167
    })
    validation: Dataset({
        features: ['image', 'label'],
        num_rows: 50000
    })
    test: Dataset({
        features: ['image', 'label'],
        num_rows: 100000
    })
})
```

6 AlexNet

AlexNet (2012): CNN para ImageNet que inició la revolución del aprendizaje profundo en el ILSVRC 2012



7 Dropout

Dropout: capa de regularización de las salidas de otra capa (entradas para dropout)

- Entrenamiento: anula cada entrada con una probabilidad dada p; las no anuladas se re-escalan con $\dfrac{1}{1-p}$
- Evaluación: actúa como la función identidad

Ejemplo: dropout aplicado a $\mathbf{X} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ en los siguientes casos

1. $p = 0.1,$ asumiendo que no anula ninguna entrada: $\mathrm{Dropout}(\mathbf{X}, 0.1) = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \dfrac{1}{0.9} = \begin{pmatrix} 1.1111 \\ 4.4444 \end{pmatrix}$

2. $p = 0.5,$ asumiendo que anula la primera entrada: $\mathrm{Dropout}(\mathbf{X}, 0.5) = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \dfrac{1}{0.5} = \begin{pmatrix} 0 \\ 8 \end{pmatrix}$

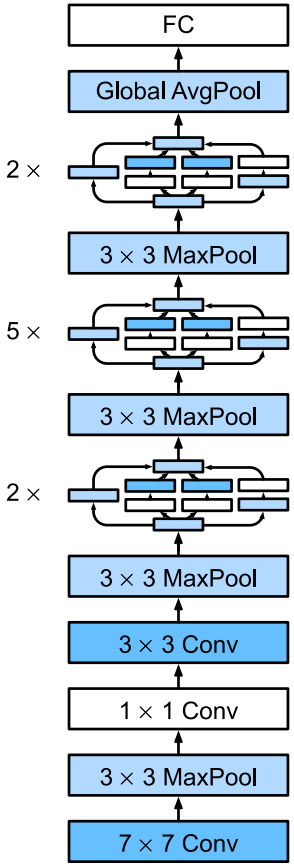
In [ ]:
```python
import torch; torch.random.manual_seed(23); X = torch.tensor([1, 4], dtype=torch.float)
print(torch.nn.Dropout(p=0.1)(X), torch.nn.Dropout(p=0.5)(X))
```

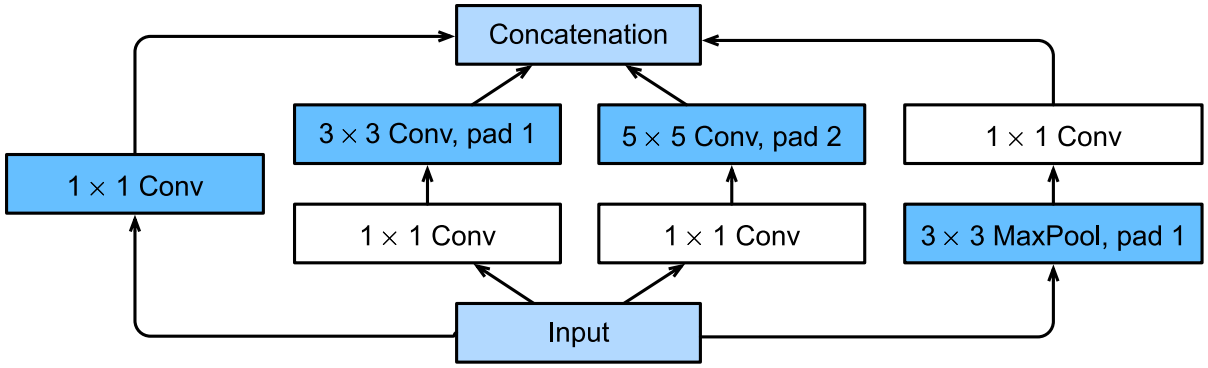
```
tensor([1.1111, 4.4444]) tensor([0., 8.])
```

8 Inception

Inception (2014): CNN de Google que introduce la meta-arquitectura típica de las CNNs modernas

- **Stem (raíz):** dos o tres capas convolucionales que extraen características de bajo nivel
- **Body (cuerpo):** subred de **bloques convolucionales** repetidos
- **Head (cabeza):** transforma la salida del cuerpo según la tarea a abordar (clasificación, segmentación, etc.)





Inception v3 (2015): Rethinking the Inception Architecture for Computer Vision

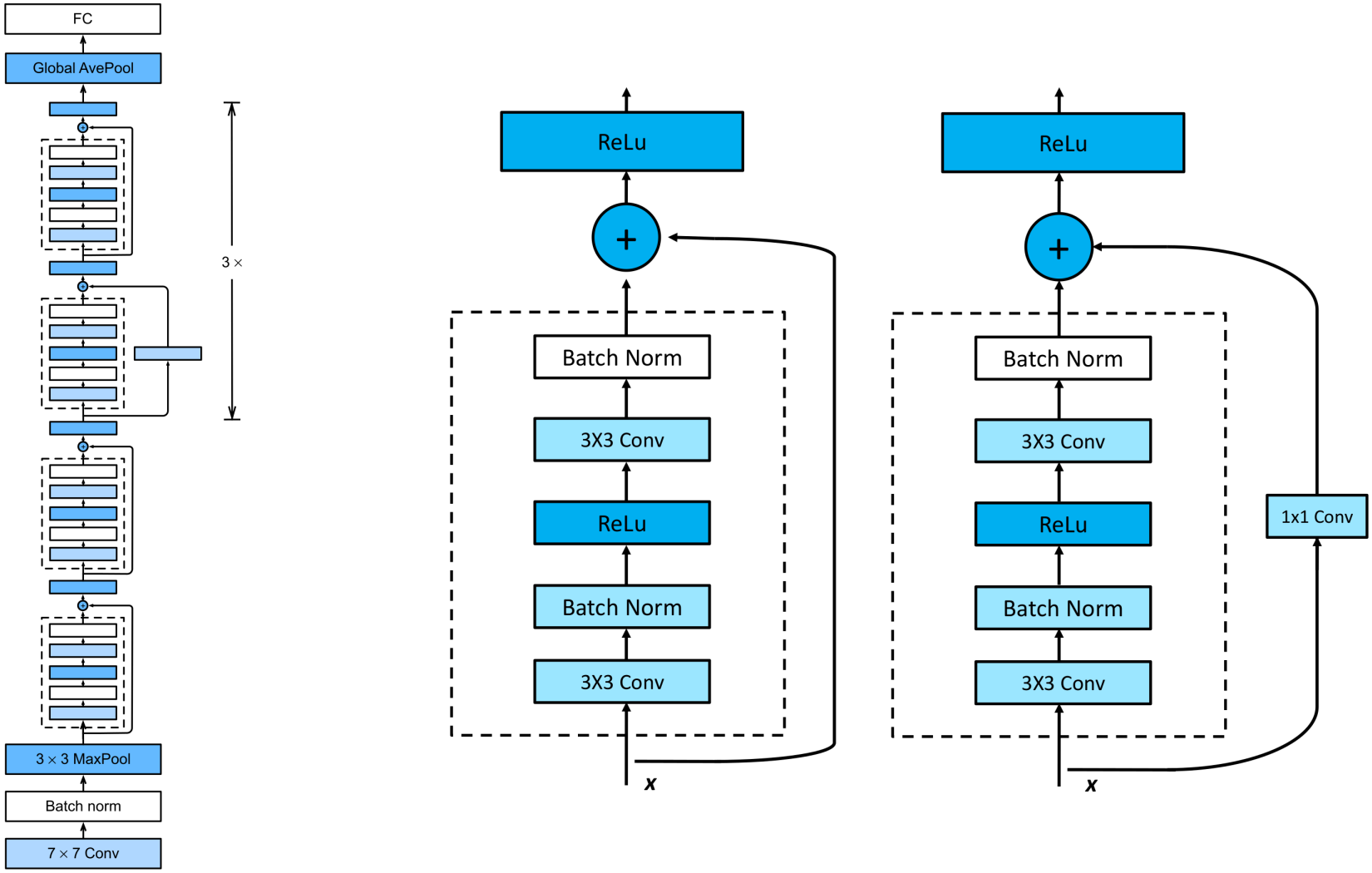
In [ ]:
```
import matplotlib.pyplot as plt; import urllib; from PIL import Image; import torch
import timm; from timm.data import resolve_data_config; from timm.data.transforms_factory import create_transform
model = timm.create_model('inception_v3', pretrained=True); model.eval()
config = resolve_data_config({}, model=model); transform = create_transform(**config)
url, file = ("https://github.com/pytorch/hub/raw/master/images/dog.jpg", "dog.jpg")
urllib.request.urlretrieve(url, file); img = Image.open(file).convert('RGB')
tensor = transform(img).unsqueeze(0); plt.xticks([]); plt.yticks([]); plt.imshow(img);
```

In [ ]:
```
probs = torch.nn.functional.softmax(model(tensor)[0], dim=0)
url, file = ("https://raw.githubusercontent.com/pytorch/hub/master/imagenet_classes.txt", "imagenet_classes.txt")
urllib.request.urlretrieve(url, file); cats = [s.strip() for s in open("imagenet_classes.txt", "r").readlines()]
i = probs.argmax().item(); print(f'{cats[i]} {probs[i].item():.2%} ')
```

```
Samoyed 83.71%
```

9 ResNet

ResNet (2015): CNN de Microsoft que introduce **conexiones residuales** para entrenar modelos muy profundos



ResNet (2015): Deep Residual Learning for Image Recognition

In [ ]:
```
import matplotlib.pyplot as plt; import urllib; from PIL import Image; import torch
import timm; from timm.data import resolve_data_config; from timm.data.transforms_factory import create_transform
model = timm.create_model('resnet18', pretrained=True); model.eval()
config = resolve_data_config({}, model=model); transform = create_transform(**config)
url, file = ("https://github.com/pytorch/hub/raw/master/images/dog.jpg", "dog.jpg")
urllib.request.urlretrieve(url, file); img = Image.open(file).convert('RGB')
tensor = transform(img).unsqueeze(0); plt.xticks([]); plt.yticks([]); plt.imshow(img);
```

In [ ]:
```
probs = torch.nn.functional.softmax(model(tensor)[0], dim=0)
url, file = ("https://raw.githubusercontent.com/pytorch/hub/master/imagenet_classes.txt", "imagenet_classes.txt")
urllib.request.urlretrieve(url, file); cats = [s.strip() for s in open("imagenet_classes.txt", "r").readlines()]
i = probs.argmax().item(); print(f'{cats[i]} {probs[i].item():.2%} ')
```

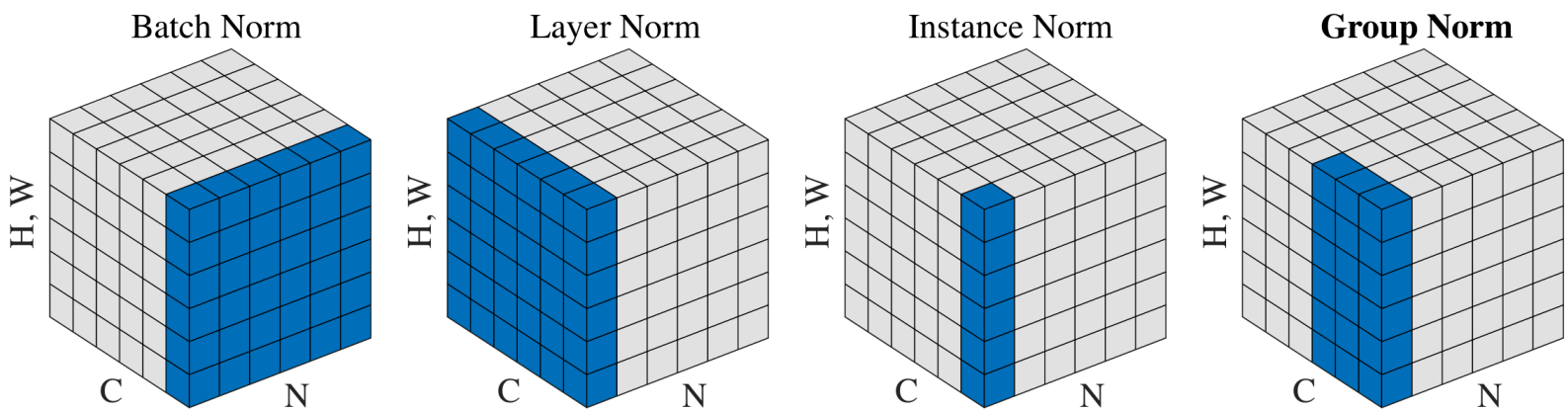
```
Samoyed 99.96%
```

10 Capas de normalización

Objetivo: añadir capas extra para evitar el problema de los gradientes que desaparecen o explotan en redes profundas

Batch/layer/instance/group norm: $\gamma = 1$ y $\beta = 0$ pueden aprenderse; $\epsilon = 10^{-5}$ por omisión

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}}\,\gamma + \beta$$



BatchNorm (2015): Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift

InstanceNorm (2016): Instance Normalization: The Missing Ingredient for Fast Stylization

LayerNorm (2016): Layer Normalization

GroupNorm (2018): Group Normalization

Ejemplo batch norm: entrada $\mathbf{X}\, N \times C \times x_h \times x_w$ y salida $\mathbf{Y}\, N \times C \times x_h \times x_w$

$$\mathbf{X} = [[[0],[1]],[[2],[3]]]$$

$$\mathbf{Y} = \mathrm{BatchNorm2d}(\mathbf{X}) = [[[-1],[-1]],[[1],[1]]]$$

In [ ]:
```
import torch; X = torch.tensor([ [ [[0]], [[1]] ], [ [[2]], [[3]] ] ], dtype=float)
print(X.shape, str(X).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 2, 1, 1]) tensor([[[[0.]],[[1.]]],[[[2.]],[[3.]]]],dtype=torch.float64)
```

In [ ]:
```
Y = torch.nn.BatchNorm2d(num_features=2, eps=0., momentum=0., affine=False, dtype=X.dtype)(X)
print(Y.shape, str(Y).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 2, 1, 1]) tensor([[[[-1.]],[[-1.]]],[[[1.]],[[1.]]]],dtype=torch.float64)
```

Paso a paso:

In [ ]:
```
X_flat = X.transpose(0, 1).flatten(start_dim=1)
print(X_flat.shape, str(X_flat).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 2]) tensor([[0.,2.],[1.,3.]],dtype=torch.float64)
```

In [ ]:
```
X_mean = X_flat.mean(dim=1).reshape(-1, 1)
print(X_mean.shape, str(X_mean).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 1]) tensor([[1.],[2.]],dtype=torch.float64)
```

In [ ]:
```
X_std = X_flat.std(dim=1, correction=0).reshape(-1, 1)
print(X_std.shape, str(X_std).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 1]) tensor([[1.],[1.]],dtype=torch.float64)
```

In [ ]:
```
Y_alt = ((X_flat - X_mean) / X_std).transpose(0, 1).reshape(X.shape); (Y - Y_alt).abs().max().item()
```

Out[ ]:
```
0.0
```

Ejemplo layer norm: entrada $\mathbf{X}\, N \times C \times x_h \times x_w$ y salida $\mathbf{Y}\, N \times C \times x_h \times x_w$

$$\mathbf{X} = \left[\left[\begin{bmatrix} 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right] \right]$$

$$\mathbf{Y} = \mathrm{LayerNorm}(\mathbf{X}) = \left[\left[\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right] \right]$$

In [ ]:
```
import torch; X = torch.tensor([ [ [[0], [2]], [[2], [0]] ] ], dtype=float)
print(X.shape, str(X).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 2, 2, 1]) tensor([[[[0.],[2.]],[[2.],[0.]]]],dtype=torch.float64)
```

In [ ]:
```
Y = torch.nn.LayerNorm(X.shape[1:], eps=0., elementwise_affine=False, bias=False, dtype=X.dtype)(X)
print(Y.shape, str(Y).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 2, 2, 1]) tensor([[[[-1.],[1.]],[[1.],[-1.]]]],dtype=torch.float64)
```

Paso a paso:

In [ ]:
```
X_flat = X.flatten(start_dim=1)
print(X_flat.shape, str(X_flat).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 4]) tensor([[0.,2.,2.,0.]],dtype=torch.float64)
```

In [ ]:
```
X_mean = X_flat.mean(dim=1).reshape(-1, 1)
print(X_mean.shape, str(X_mean).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1]) tensor([[1.]],dtype=torch.float64)
```

In [ ]:
```
X_std = X_flat.std(dim=1, correction=0).reshape(-1, 1)
print(X_std.shape, str(X_std).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1]) tensor([[1.]],dtype=torch.float64)
```

In [ ]:
```
Y_alt = ((X_flat - X_mean) / X_std).reshape(X.shape); (Y - Y_alt).abs().max().item()
```

Out[ ]:
```
0.0
```

Ejemplo instance norm: entrada $\mathbf{X}\, N \times C \times x_h \times x_w$ y salida $\mathbf{Y}\, N \times C \times x_h \times x_w$

$$\mathbf{X} = \left[\left[\left[\begin{matrix} 0 & 2 \\ 2 & 0 \end{matrix}\right]\right]\right]$$

$$\mathbf{Y} = \mathrm{InstanceNorm2d}(\mathbf{X}) = \left[\left[\left[\begin{matrix} -1 & 1 \\ 1 & -1 \end{matrix}\right]\right]\right]$$

In [ ]:
```
import torch; X = torch.tensor([ [ [ [0, 2], [2, 0] ] ] ], dtype=float)
print(X.shape, str(X).replace('\n',''))
```

```
torch.Size([1, 1, 2, 2]) tensor([[[[0., 2.],          [2., 0.]]]], dtype=torch.float64)
```

In [ ]:
```
Y = torch.nn.InstanceNorm2d(num_features=1, eps=0., momentum=0., affine=False, dtype=X.dtype)(X)
print(Y.shape, str(Y).replace('\n',''))
```

```
torch.Size([1, 1, 2, 2]) tensor([[[[-1.,  1.],          [ 1., -1.]]]], dtype=torch.float64)
```

Paso a paso:

In [ ]:
```
X_flat = X.flatten(start_dim=2)
print(X_flat.shape, str(X_flat).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1, 4]) tensor([[[0.,2.,2.,0.]]],dtype=torch.float64)
```

In [ ]:
```
X_mean = X_flat.mean(dim=2).reshape(-1, 1)
print(X_mean.shape, str(X_mean).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1]) tensor([[1.]],dtype=torch.float64)
```

In [ ]:
```
X_std = X_flat.std(dim=2, correction=0).reshape(-1, 1)
print(X_std.shape, str(X_std).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1]) tensor([[1.]],dtype=torch.float64)
```

In [ ]:
```
Y_alt = ((X_flat - X_mean) / X_std).reshape(X.shape); (Y - Y_alt).abs().max().item()
```

Out[ ]:
```
0.0
```

Ejemplo group norm: entrada $\mathbf{X}\, N \times C \times x_h \times x_w$ y salida $\mathbf{Y}\, N \times C \times x_h \times x_w$

$$\mathbf{X} = [[[0],[1],[2],[3]]]$$

$$\mathbf{Y} = \mathrm{GroupNorm}(\mathbf{X}, 2) = [[[-1],[1],[-1],[1]]]$$

In [ ]:
```
import torch; X = torch.tensor([ [ [[0]], [[1]], [[2]], [[3]] ] ], dtype=float)
print(X.shape, str(X).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 4, 1, 1]) tensor([[[[0.]],[[1.]],[[2.]],[[3.]]]],dtype=torch.float64)
```

In [ ]:
```
Y = torch.nn.GroupNorm(num_groups=2, num_channels=4, eps=0., affine=False, dtype=X.dtype)(X)
print(Y.shape, str(Y).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 4, 1, 1]) tensor([[[[-1.]],[[1.]],[[-1.]],[[1.]]]],dtype=torch.float64)
```

Paso a paso:

In [ ]:
```
X_flat = X.reshape(X.shape[0], 2, X.shape[1]//2, *X.shape[2:]).flatten(start_dim=2)
print(X_flat.shape, str(X_flat).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 2, 2]) tensor([[[0.,1.],[2.,3.]]],dtype=torch.float64)
```

In [ ]:
```
X_mean = X_flat.mean(dim=2).reshape(-1, 1)
print(X_mean.shape, str(X_mean).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 1]) tensor([[0.5000],[2.5000]],dtype=torch.float64)
```

In [ ]:
```
X_std = X_flat.std(dim=2, correction=0).reshape(-1, 1)
print(X_std.shape, str(X_std).replace('\n','').replace(' ',''))
```

```
torch.Size([2, 1]) tensor([[0.5000],[0.5000]],dtype=torch.float64)
```

In [ ]:
```
Y_alt = ((X_flat - X_mean) / X_std).reshape(X.shape); (Y - Y_alt).abs().max().item()
```

Out[ ]:
```
0.0
```

RMSNorm: Root Mean Square Layer Normalization se aplica sobre las últimas dimensiones del batch

$$y_i = \frac{x_i}{\mathrm{RMS}(x)} * \gamma_i, \quad \text{donde} \quad \mathrm{RMS}(x) = \sqrt{\epsilon + \frac{1}{n}\sum_{i=1}^{n} x_i^2}$$

Ejemplo: entrada $\mathbf{X}\, N \times C \times x_h \times x_w$ y salida $\mathbf{Y}\, N \times C \times x_h \times x_w$ sobre la última dimensión

$$\mathbf{X} = \left[\left[\left[\begin{matrix} 1 & 1 \\ 2 & 2 \end{matrix}\right]\right]\right] \qquad \mathbf{Y} = \mathrm{RMSNorm}(\mathbf{X}, (2,)) = \left[\left[\left[\begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix}\right]\right]\right]$$

In [ ]:
```
import torch; X = torch.tensor([ [ [ [1, 1], [2, 2] ] ] ], dtype=float)
print(X.shape, str(X).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1, 2, 2]) tensor([[[[1.,1.],[2.,2.]]]],dtype=torch.float64)
```

In [ ]:
```
Y = torch.nn.RMSNorm(normalized_shape=X.shape[-1:], eps=0., elementwise_affine=False, dtype=X.dtype)(X)
print(Y.shape, str(Y).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1, 2, 2]) tensor([[[[1.,1.],[1.,1.]]]],dtype=torch.float64)
```

Paso a paso:

In [ ]:
```
X_flat = X.flatten(start_dim=-1)
print(X_flat.shape, str(X_flat).replace('\n','').replace(' ',''))
```

```
torch.Size([1, 1, 2, 2]) tensor([[[[1.,1.],[2.,2.]]]],dtype=torch.float64)
```

In [ ]:
```
X_rms = X_flat.square().mean(dim=-1).reshape(-1, 1).sqrt()
print(X_rms.shape, str(X_rms).replace('\n','').replace(' ',''))
```

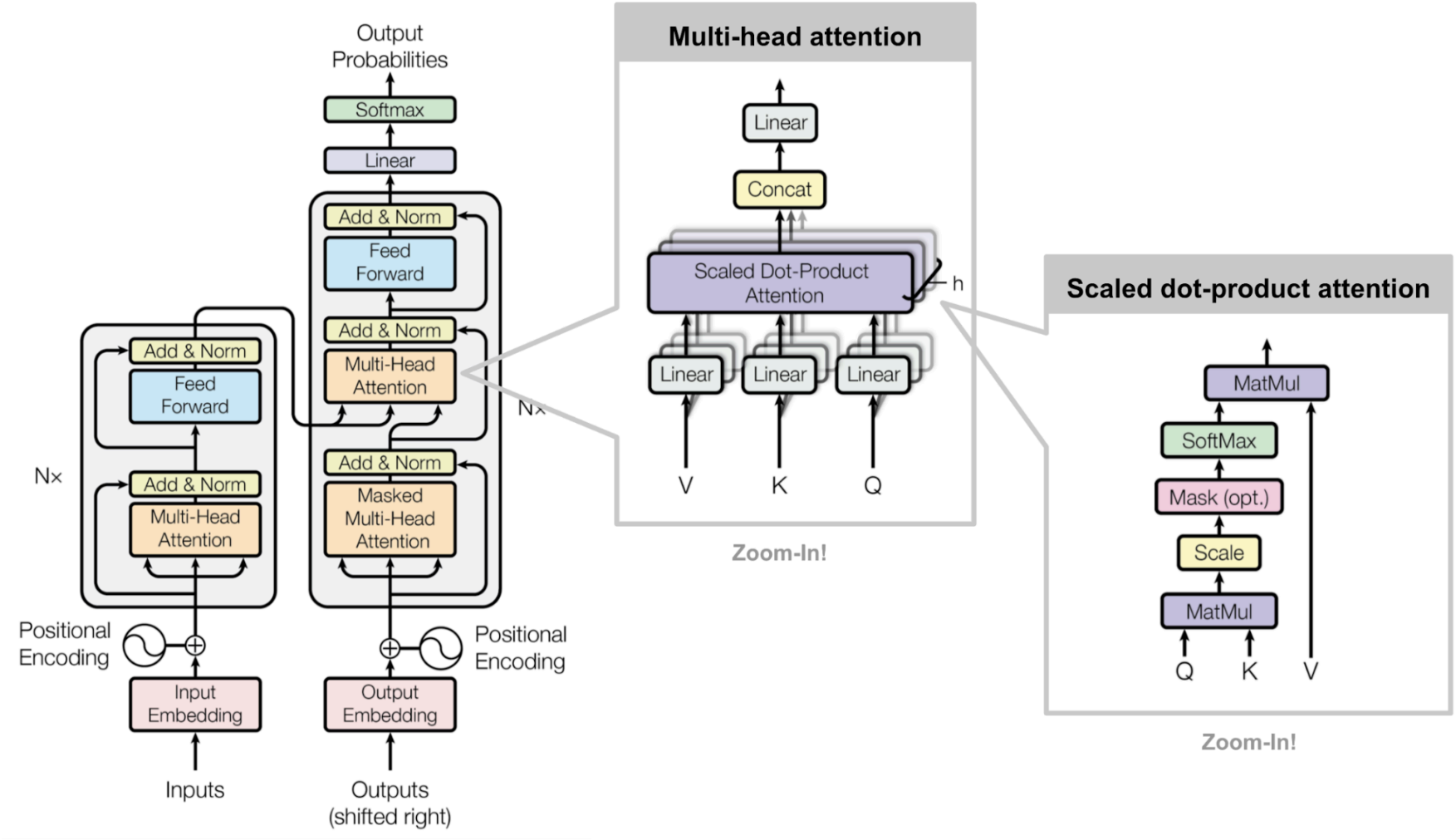
```
torch.Size([2, 1]) tensor([[1.],[2.]],dtype=torch.float64)
```

In [ ]:
```
Y_alt = (X_flat / X_rms).reshape(X.shape); (Y - Y_alt).abs().max().item()
```

Out[ ]:
```
0.0
```

11 Transformer

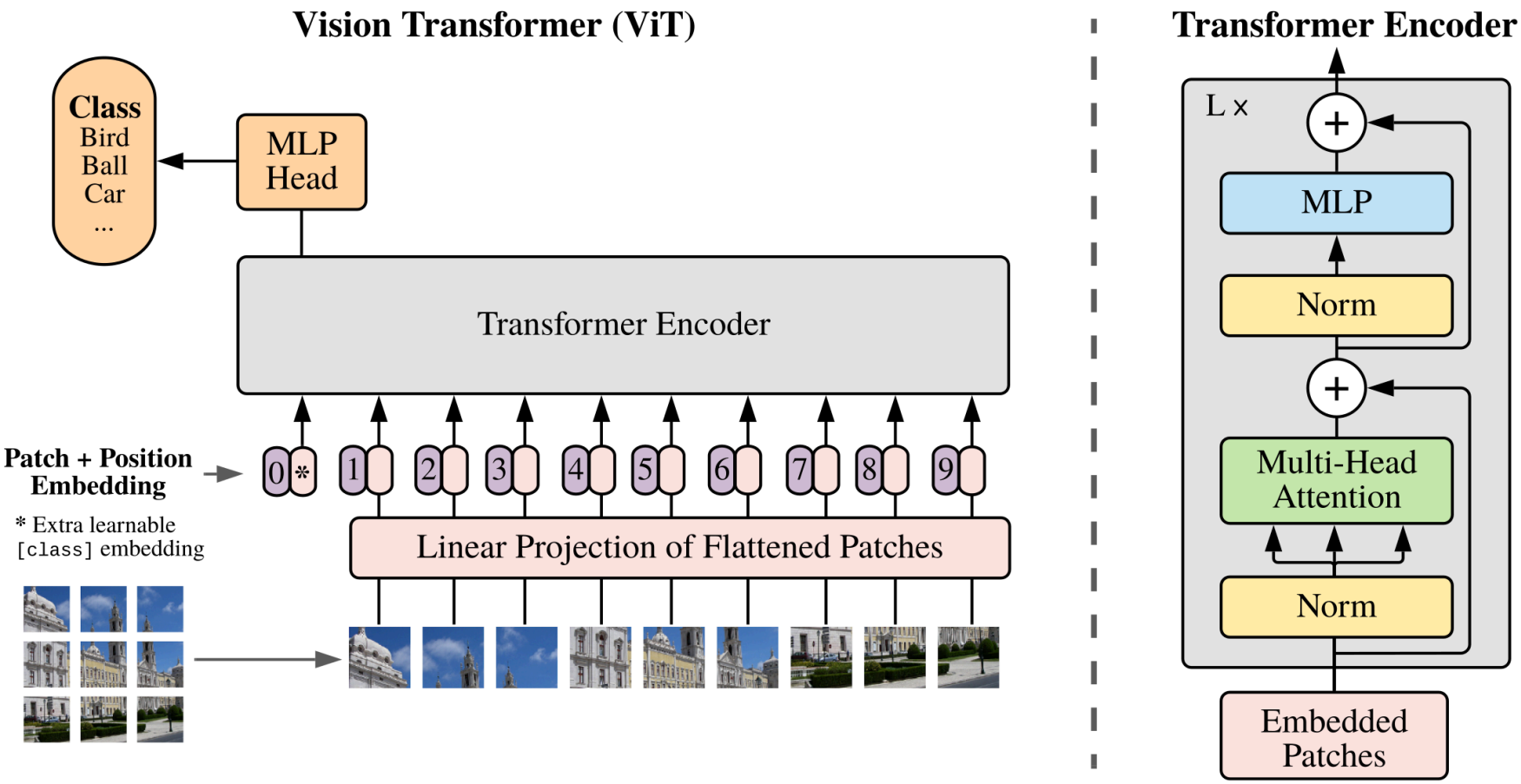
Transformer (2017): arquitectura de Google para traducción automática basada en el **mecanismo de atención**



3blue1brown: ver Transformers, the tech behind LLMs y Attention in transformers, step-by-step

12 Vision Transformer

Vision Transformer (2020): adaptación del Transformer para visión que rivaliza con las CNNs



SOTA en ImageNet: ver ficheros de resultados de Ross Wightman